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First Mock Periodic Test (2026–2027)

Class: X

Subject: Mathematics (041)

Chapter: Real Numbers

Maximum Marks: 20

Time Allowed: 45 Minutes

General Instructions:

1. This question paper contains 8 questions divided into four sections: Section A, Section B, Section C, and Section D.
2. Section A comprises Multiple Choice Questions (MCQs) of 1 mark each.
3. Section B comprises Short Answer Type-I (SA-I) questions of 2 marks each.
4. Section C comprises Short Answer Type-II (SA-II) questions of 3 marks each.
5. Section D comprises a Long Answer (LA) question of 5 marks.
6. Use of calculators is strictly prohibited. Show all necessary working clearly.

SECTION A (1 Mark each)

Q1. The exponent of 2 in the prime factorisation of 144 is:

- (a) 2
- (b) 3
- (c) 4
- (d) 5

Q2. Given that $\text{HCF}(306, 657) = 9$, the $\text{LCM}(306, 657)$ is:

- (a) 22338
- (b) 22238
- (c) 21933
- (d) 23322

SECTION B (2 Marks each)

Q3. Express the number 3825 as a product of its prime factors.

Q4. Explain why $7 \times 11 \times 13 + 13$ is a composite number.

Q5. Find the HCF and LCM of 6 and 20 by the prime factorisation method.

SECTION C (3 Marks each)

Q6. Check whether 6^n can end with the digit 0 for any natural number n .

Q7. Find the LCM and HCF of the integers 12, 15 and 21 by applying the prime factorisation method.

SECTION D (5 Marks)

Q8. Prove that the $\sqrt{5}$ is an irrational number.

ANSWER KEY

Question No.	Answer Key
Q1	(c) 4 (Hint: $144 = 16 \times 9 = 2^4 \times 3^2$, so the exponent of 2 is 4)
Q2	(a) 22338 ($\text{LCM} = (306 \times 657) / 9 = 306 \times 73 = 22338$)
Q3	$3825 = 3^2 \times 5^2 \times 17$ (or $3 \times 3 \times 5 \times 5 \times 17$)
Q4	Taking 13 common: $13 \times (7 \times 11 \times 1 + 1) = 13 \times (77 + 1) = 13 \times 78$. Since the number has factors other than 1 and itself, it is a composite number.
Q5	$\text{HCF}(6, 20) = 2$; $\text{LCM}(6, 20) = 60$
Q6	If 6^n ends with 0, it must be divisible by 5. The prime factorisation of 6^n is $(2 \times 3)^n = 2^n \times 3^n$. The only prime factors are 2 and 3. By the uniqueness of the Fundamental Theorem of Arithmetic, 5 is not a factor. Thus, 6^n cannot end with 0 for any natural number n .
Q7	Prime factorisations: $12 = 2^2 \times 3$; $15 = 3 \times 5$; $21 = 3 \times 7$; $\text{HCF}(12, 15, 21) = 3$; $\text{LCM}(12, 15, 21) = 2^2 \times 3 \times 5 \times 7 = 420$
Q8	Proof: Assume root 5 is rational, i.e., $\sqrt{5} = a/b$. Then $5 = a^2/b^2$, or $a^2 = 5b^2$. This means 5 divides a^2 , and thus 5 divides a . Letting $a = 5c$, we get $(5c)^2 = 5b^2$, so $25c^2 = 5b^2$, or $b^2 = 5c^2$. This means 5 divides b . Both a and b have a common factor 5, contradicting the coprimality of a and b . Thus, root 5 is irrational.